Project Ideology & Base Paper Deep Study

AI-Based E-Commerce Product Representation Consistency and Reliability Assessment System

PART A: OUR PROJECT IDEOLOGY & DESIGN APPROACH

A1. The Problem We Are Solving

A1.1 The Disconnect in Online Shopping

Every online purchase begins with a trust gap. The buyer cannot touch, feel, measure, or test the product. They depend entirely on the seller's digital representation: titles, bullet points, specifications, and studio-lit catalog photos. After delivery, the reality can differ: the "leather" feels like coated plastic, shoes run two sizes smaller, colors look nothing like the catalog image, or the "non-slip sole" is smooth and slippery.

A1.2 Why Current Solutions Fail

Star Ratings (e.g., 4.3/5.0): Collapse the entire customer experience into a single number. A customer who writes "Love the design! 5 stars! But this is definitely polyester, not cotton" contributes a perfect 5-star rating. The material contradiction is invisible in the aggregated number.
Sentiment Analysis: Measures the emotion of the reviewer (positive/negative), not the accuracy of the seller's claims. A positive review warning about a false specification gets classified as "Positive" and the warning is lost.
Fake Review Detection: Asks "Was this review written by a bot?" Even if every review is authentic, nobody checks whether the seller's actual product claims match what those real customers experienced.

A1.3 Our Core Question

"For each specific claim the seller makes - about material, sizing, color, function - does the available body of customer evidence support, contradict, or remain ambiguous about that claim?"

A1.4 Our Philosophy: Reliability Engineering, Not Fraud Detection

We deliberately avoid the words "fraud", "fake", or "lying". Intent is unprovable in code. A seller might have received a defective batch from a manufacturer, sizing standards might differ between countries, or studio lighting might exaggerate colors. Our system is a Quality Assurance (QA) audit. We report empirical consistency: "The claim 'Genuine Leather' has 86% contradiction across 42 verified customer reviews." This is objective, measurable, and defensible.

A2. All Possible Ways to Solve This Problem

Before designing our solution, we explore every possible technical approach honestly, with strengths and weaknesses.

Approach 1: Rule-Based / Keyword Matching

How it works: Write manual rules ("if seller says 'leather' AND review contains 'plastic', flag contradiction").
Strengths: Extremely simple, no AI models, no GPU required, easy to explain.
Weaknesses: Fails on indirect language ("This material won't survive a week" implies poor quality but has no keyword match). Fails on sarcasm. Requires manually writing hundreds of rules per category. Cannot scale.
Verdict: Too brittle. Works for a toy demo but not for a real system or academic project.

Approach 2: Classical ML (TF-IDF + Random Forest / SVM)

How it works: Extract hand-crafted features (word counts, TF-IDF vectors, sentiment scores, keyword overlap), then train a classifier (Random Forest, SVM) to predict "Consistent" vs. "Contradictory".
Strengths: Well-understood algorithms, fast, low compute.
Weaknesses: TF-IDF treats words as isolated symbols and cannot understand that "runs small" and "order a size up" mean the same thing. Requires a large manually-labeled training dataset (thousands of annotated claim-review pairs). Creating such a dataset is massive manual effort for a student team.
Verdict: Academically outdated for NLP tasks in 2025. Would look weak against the faculty's other titles.

Approach 3: Fine-Tuned BERT/RoBERTa

How it works: Take a pretrained language model (BERT/RoBERTa), fine-tune it on a custom labeled dataset of (claim, review, label) triplets where labels are Support/Contradict/Neutral.
Strengths: Understands context, synonyms, and indirect language. State-of-the-art accuracy. Generalizes well.
Weaknesses: Still requires a labeled training dataset (which we would need to create). Fine-tuning requires GPU access. Risk of overfitting on a small custom dataset.
Verdict: Strong approach, but we can avoid the dataset creation burden entirely by using pretrained NLI models (see Approach 4).

Approach 4: Zero-Shot / Pretrained NLI (No Custom Training)

How it works: Use a model like cross-encoder/nli-roberta-base that is already trained on 570K+ sentence pairs for NLI. Feed it (Seller Claim, Review Sentence) and it outputs Support/Contradict/Neutral probabilities without any training.
Strengths: No custom dataset creation. No fine-tuning or GPU training. Runs on a standard laptop CPU. Deterministic and reproducible results. Academically defensible.
Weaknesses: Slightly less accurate than a model fine-tuned specifically on e-commerce data. Requires a smart retrieval layer to avoid feeding irrelevant sentences.
Verdict: Strong candidate for the text inference engine.

Approach 5: LLM Prompting (Like the PRAISE Base Paper)

How it works: Send structured prompts to a large language model (Gemini, GPT-4) asking it to extract attributes from reviews and compare them with descriptions. This is exactly what the PRAISE paper does.
Strengths: Extremely flexible, handles any category, can extract structured JSON output in one pass.
Weaknesses: Cost (PRAISE makes ~2R+1 API calls per product, 200+ calls for 100 reviews). Non-determinism (LLMs give slightly different outputs on repeated runs). Reproducibility concerns for academic evaluation. Latency (each call takes 1-3s). Dependency on API provider pricing and model changes.
Verdict: Powerful for attribute extraction, but risky as the sole backbone.

Approach 6: Hybrid Architecture (Multi-Stage Pipeline)

How it works: Use different tools for different stages, matching each tool to its strength:
- Stage 1 (Claim Extraction): LLM or rule-based parser to break seller text into atomic claims
- Stage 2 (Retrieval): Fast Bi-Encoder (Sentence-BERT) to find relevant review sentences per claim
- Stage 3 (Inference): Pretrained NLI Cross-Encoder (RoBERTa) to classify Support/Contradict
- Stage 4 (Visual, if applicable): Computer vision to compare catalog vs. customer photos

Strengths: Each component uses the best tool for its task. Runs locally without API costs. Deterministic. Modular.
Weaknesses: More components to integrate. Requires understanding multiple models.
Verdict: Most robust, scalable, and academically defensible architecture.

A3. Comparison Summary of All Approaches

Approach	Accuracy	Training Data	Cost	Reproducible	Academic Level	Verdict
1. Rule-Based	Low	None	Free	Yes	Weak	Too brittle
2. Classical ML	Medium	Large labeled set	Free	Yes	Outdated 2025	Not competitive
3. Fine-Tuned DL	High	Custom labeled	GPU time	Yes	Strong	Dataset burden
4. Pretrained NLI	High	None	Free	Yes	Strong	Strong candidate
5. LLM Prompting	Very High	None	API costs	Partial	Very Strong	Cost/determinism risk
6. Hybrid	Very High	None	Free	Yes	Very Strong	Best balance

NOTE: The final choice of approach is open for your confirmation. Based on the analysis, the Hybrid Architecture (Approach 6) offers the best balance of accuracy, cost, reproducibility, and academic rigor. But every architectural decision will be finalized through collaborative discussion.

PART B: DEEP STUDY OF THE BASE PAPER - PRAISE (ACL 2025)

B1. Paper Identity

Title: PRAISE: Enhancing Product Descriptions with LLM-Driven Structured Insights
Authors: Adnan Qidwai*, Srija Mukhopadhyay* (IIIT Hyderabad), Prerana Khatiwada (Univ. of Delaware), Dan Roth (Univ. of Pennsylvania), Vivek Gupta (Arizona State University)
Venue: ACL 2025 (63rd Annual Meeting of the Association for Computational Linguistics), Volume 3: System Demonstrations, pp. 644-652. July-August 2025, Vienna, Austria.
Conference Rank: Core A* - the single most prestigious NLP conference in the world.

Who Are the Authors?

IIIT Hyderabad: One of India's top-ranked research institutions for NLP and AI. University of Pennsylvania (UPenn): Dan Roth is a world-renowned NLP researcher, former VP/Distinguished Scientist at Amazon AI. Arizona State University (ASU): Vivek Gupta leads research in structured reasoning and NLP. The asterisk (*) on the first two authors indicates they contributed equally as co-first authors.

What Does "ACL" Mean?

ACL stands for the Association for Computational Linguistics. Its annual meeting is universally recognized as the #1 venue for NLP research globally. Papers undergo rigorous peer review by world-class researchers. Having an ACL 2025 paper as your base paper is the strongest possible academic anchor for a student project.

B2. The Problem PRAISE Solves

Consider a product listing for wireless earbuds. The seller claims: "Active Noise Cancellation. 30-hour battery life. IPX5 waterproof. Deep bass with 13mm drivers." Across 200 customer reviews, sentences like these are buried:
- "Battery barely lasts 8 hours, nowhere near 30 claimed" (Contradicts battery claim)
- "ANC works decently, blocks out office noise" (Supports ANC claim)
- "Got it wet in rain and it died" (Contradicts waterproof claim)
- "Shipping was fast, arrived in 2 days" (Irrelevant noise)

PRAISE's Core Insight: Customer reviews collectively contain a goldmine of factual, ground-truth information about the product. But this information is: (1) Scattered across hundreds of reviews, (2) Buried under irrelevant opinions and personal stories, (3) Unstructured raw paragraphs, not organized fact sheets, and (4) Never formally compared against the seller's claims. PRAISE builds an automated system to extract these factual nuggets and compare them against what the seller stated.

B3. How PRAISE Works - The 4-Stage Pipeline

Stage 1: Attribute Extraction from Reviews

Each customer review is fed to the LLM (Gemini 2.0 Flash) with a prompt asking it to extract only objective, factual attribute-value pairs while ignoring subjective opinions and irrelevant commentary.

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INPUT (one review):
"Battery life is disappointing. Claimed 30 hours but I barely get 8 hours on a full charge. Sound quality is great though, nice deep bass. Shipping was quick."

LLM OUTPUT (structured extraction):
  attribute: "battery_life", value: "approximately 8 hours on full charge"
  attribute: "sound_quality", value: "great, deep bass"

FILTERED OUT: "disappointing" (subjective opinion), "Shipping was quick" (logistics, not product)
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The LLM is explicitly instructed to: (1) Extract facts, not feelings, (2) Ignore shipping, packaging, and customer service comments, (3) Preserve the original reviewer's language without hallucinating new claims. This stage runs once per review. For R reviews, this makes R LLM calls.

Stage 2: Cross-Comparison Against Seller Description

Each extracted attribute is compared against the official seller description. The LLM classifies the relationship into one of four categories:

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1. Matching: Review confirms the seller's claim.
   Seller: "Active Noise Cancellation" -> Review: "ANC works decently" -> MATCHING

2. Contradictory: Review directly conflicts with the seller's claim.
   Seller: "30-hour battery life" -> Review: "barely get 8 hours" -> CONTRADICTORY

3. Missing: Review mentions a fact the seller never disclosed.
   Review: "Charging case feels flimsy and cheap" -> Seller never mentioned case quality -> MISSING

4. Partially Matching: Review gives a nuanced or conditional version of the claim.
   Seller: "IPX5 waterproof" -> Review: "Survived light sweat but I would not trust it in rain" -> PARTIAL
```

This stage makes approximately R additional LLM calls (one per review's extracted attributes, batched).

Stage 3: Attribute Grouping

By this point, the system has hundreds of extracted attributes across all reviews. Many refer to the same product aspect using different words ("battery lasts 8 hours", "charge runs out by noon", "power dies quickly" all refer to Battery Life). The LLM groups semantically similar attributes into labeled clusters:
- Battery & Charging: {battery_life: 8hr, battery_life: 6hr, charging_speed: fast}
- Audio Quality: {bass: deep, clarity: good, volume: adequate}
- Build & Design: {case_quality: flimsy, earbud_weight: light}

This stage makes 1 LLM call (processing all attributes in a single batch).

Stage 4: Structured Presentation

The grouped and categorized attributes are organized into a clean, structured display showing: which seller claims are confirmed by customer evidence, which are contradicted, and what additional attributes customers mention that the seller omitted. The output is rendered as an interactive web interface.

B4. Total Computational Cost of PRAISE

The paper explicitly documents (Section 3): "For a product with R reviews, the pipeline makes approximately 2R + 1 LLM API calls."

Breakdown:
- R calls for Stage 1 (extracting attributes from each review)
- R calls for Stage 2 (comparing each review's attributes against the description)
- 1 call for Stage 3 (grouping all attributes together)
- Total: 2R + 1

Real-world implication: A popular product with 100 reviews requires 201 separate LLM API calls.

B5. Dataset Used by PRAISE

PRAISE evaluates on the Amazon Reviews 2023 dataset (McAuley Lab, UCSD). Categories tested: All Beauty, Clothing Shoes & Jewelry, Electronics, Home & Kitchen, Sports & Outdoors. For each product, the dataset contains: title, description (seller-written paragraphs), features (bullet points), images (catalog URLs), and reviews (customer text, rating, verified_purchase status, and customer-uploaded photo URLs).

B6. How PRAISE Was Evaluated

Human Evaluation: Annotators judged PRAISE output across three criteria:
1. Accuracy: Are extracted attributes factually correct? (Not hallucinated by the LLM)
2. Relevance: Are attributes genuinely about the product? (Not about shipping or packaging)
3. Categorization Correctness: Are Matching/Contradictory/Missing labels assigned correctly?

Key Results: Attribute extraction accuracy was high (LLM reliably extracted factual statements). Categorization accuracy was strong (system correctly identified contradictions in the majority of tested cases). PRAISE outperformed simpler extractive summarization and keyword-matching baselines.

B7. Limitations Explicitly Stated by the PRAISE Authors

Limitation 1 - Text-Only Modality (Section 8):
"While PRAISE demonstrates strong performance... it is limited by its exclusive reliance on customer reviews." The system has zero image analysis capability. It cannot detect color mismatches, texture differences, or visual discrepancies between catalog photos and actual delivered product appearance.

Limitation 2 - Computational Cost at Scale (Section 3):
The 2R+1 API call overhead means processing becomes slow and expensive for products with hundreds of reviews. The authors acknowledge this as a scalability concern.

Limitation 3 - No Quantitative Reliability Score:
PRAISE outputs structured tables and lists but never produces a single number representing "How reliable is this product representation overall?" A platform analyst cannot quickly compare two products without reading the entire table.

Limitation 4 - Equal Treatment of All Reviews:
PRAISE treats every review with equal weight. It does not distinguish between a verified purchaser who used the product for 3 months vs. an unverified account with a 1-line review vs. a review with 200 helpful upvotes vs. zero engagement.

B8. How PRAISE Connects to Our Project

Our project title is: "AI-Based E-Commerce Product Representation Consistency and Reliability Assessment System."

What we inherit from PRAISE: The core paradigm of extracting factual attributes from customer reviews and systematically comparing them against seller descriptions to identify matches, contradictions, and omissions.

Where our project potentially extends beyond PRAISE: The paper's documented limitations (text-only modality, computational cost, lack of unified scoring, equal review weighting) represent potential gaps that our system could address. The specific extensions, methodology choices, and architectural decisions will be finalized through our collaborative discussion.

B9. Key Technical Terms Used in the PRAISE Paper

Term	What It Means in Context
Attribute-Value Pair	A single fact extracted from a review: attribute = "battery_life", value = "8 hours"
Structured Insight	An organized, categorized comparison result (not raw text)
LLM-Driven	The pipeline uses a Large Language Model (Gemini Flash) as its core reasoning engine
Prompt Engineering	Carefully designing the text instructions sent to the LLM for reliable, structured outputs
Few-Shot Prompting	Including 2-3 worked examples inside the prompt so the LLM understands the expected format
Extraction vs. Generation	PRAISE extracts facts already in reviews; it does not fabricate new information
Cross-Source Alignment	Comparing information from Source A (seller description) against Source B (customer reviews)
System Demonstration	The ACL paper category: the authors built and presented a working prototype